Practical 5:

Aim: Write a program to implement Shortest Job First (SJF) Preemptive Scheduling for three processes and calculate the total context switches and average waiting time. The processes have burst times 10ns, 20ns, and 30ns, arriving at 0ns, 2ns, and 6ns, respectively.

Code:

kalash@KALASH: ~

GNU nano 7.2 sjf.c *

```c
#include <stdio.h>

int main() {
    int at[3] = {0,2,6};
    int bt[3] = {10,20,30};
    int rt[3], ct[3];
    int time=0, completed=0, min, pos=-1;
    int context=0, prev=-1;

    for(int i=0;i<3;i++)
        rt[i]=bt[i];

    while(completed<3){
        min=999;
        pos=-1;

        for(int i=0;i<3;i++){
            if(at[i]<=time && rt[i]>0 && rt[i]<min){
                min=rt[i];
                pos=i;
            }
        }

        if(pos==-1){
            time++;
            continue;
        }

        if(prev!=pos && prev!=-1)
            context++;

        prev=pos;
        rt[pos]--;
        time++;

        if(rt[pos]==0){
            ct[pos]=time;
            completed++;
        }
    }

    float avgWT=0;
    for(int i=0;i<3;i++){
        avgWT += (ct[i]-at[i]-bt[i]);
    }

    printf("Average Waiting Time = %.2f\n",avgWT/3);
    printf("Total Context Switches = %d\n",context);

    return 0;
}
```

Output:

```
root@KALASH:/mnt/c/Windows/System32# cd~
Command 'cd~' not found, did you mean:
  command 'cdw' from deb cdw (0.8.1-3)
  command 'cdo' from deb cdo (2.3.0-1)
  command 'cd5' from deb cd5 (0.1-4)
  command 'cdi' from deb cdo (2.3.0-1)
  command 'cde' from deb cde (0.1+git9-g551e54d-1.2)
  command 'cdb' from deb tinycdb (0.81-1)
  command 'cdp' from deb irpas (0.10-9)
Try: apt install <deb name>
root@KALASH:/mnt/c/Windows/System32# nano sjf.c
root@KALASH:/mnt/c/Windows/System32# gcc sjf.c -o sjf
root@KALASH:/mnt/c/Windows/System32# ./sjf

Process AT      BT      CT      TAT     WT
P1      0       10      10      10      0
P2      2       20      30      28      8
P3      6       30      60      54      24

Average Waiting Time = 10.67
Total Context Switches = 2
root@KALASH:/mnt/c/Windows/System32#
```